

Digital Platform for Social Welfare Services Management in Sri Lanka

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Our Team

Supervisor : **Ms. Anjalie Gamage**

Co-Supervisor : **Ms. Geethanjali Wimalaratne**



Manawadu G.M. S. T
IT22926494



Shamika A. G
IT22585080



Dissanayake D.M.V.S
IT22366818



Wickramasinghe R P Y N
IT22214584



Research Problem

Sri Lanka's welfare services rely on manual processes, requiring citizens to visit service centers, submit documents, and wait in long queues. This results in delays, errors in data collection, limited government staff, and poor accessibility, particularly for elderly individuals and residents in remote areas, reducing efficiency and service quality.



Research Objectives

Investment Plan Recommendation

Recommend financial support or investment options to users (e.g., microloans or savings plans).

Eligibility Prediction & Healthcare Recommendation

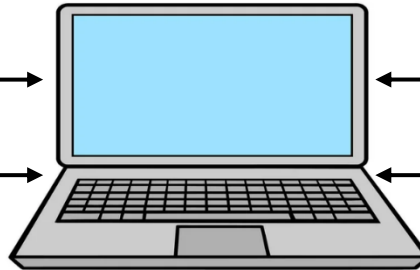
Predict who qualifies for welfare programs and suggest hospitals or doctors (e.g., for pregnant mothers or elderly patients).

AI-GIS Welfare Insights Platform

Map areas with high welfare demand to help policymakers (e.g., regions with many chronic patients or low-income families).

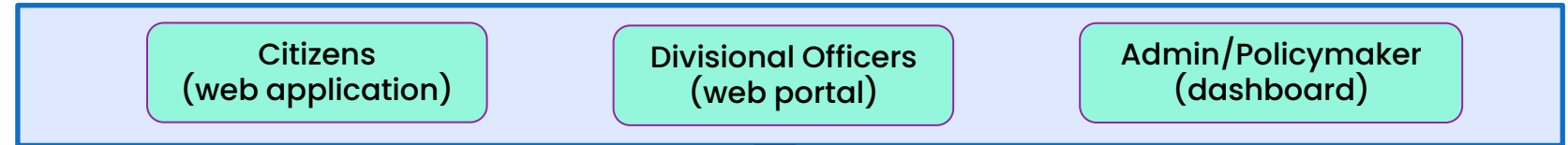
Multilingual Conversational Assistance

Build a chatbot in Sinhala, Tamil, and English to guide users on applications and documents (e.g., check application status or required forms).

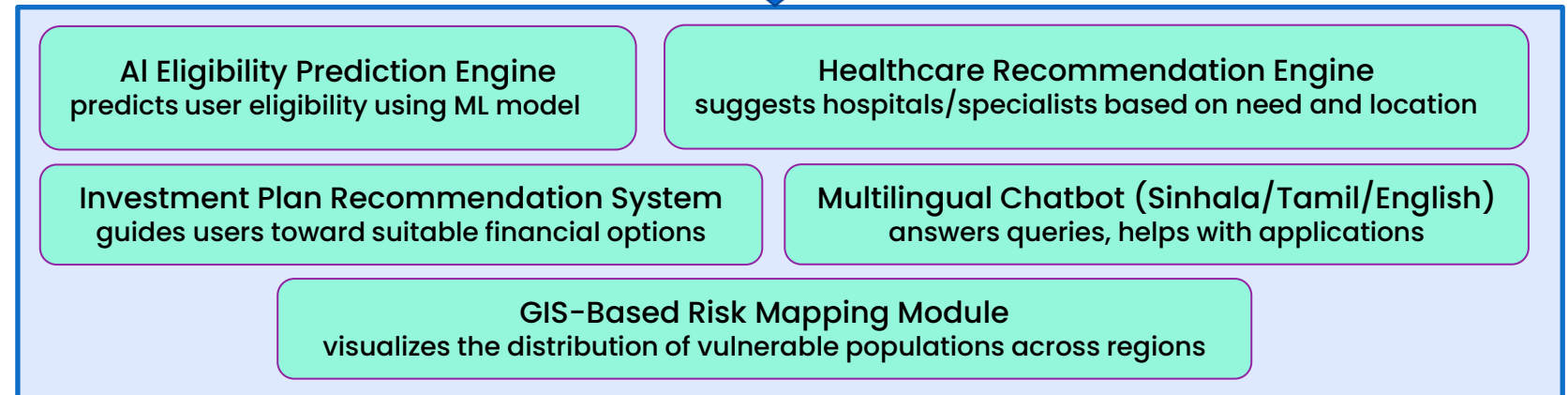


Overall System Diagram

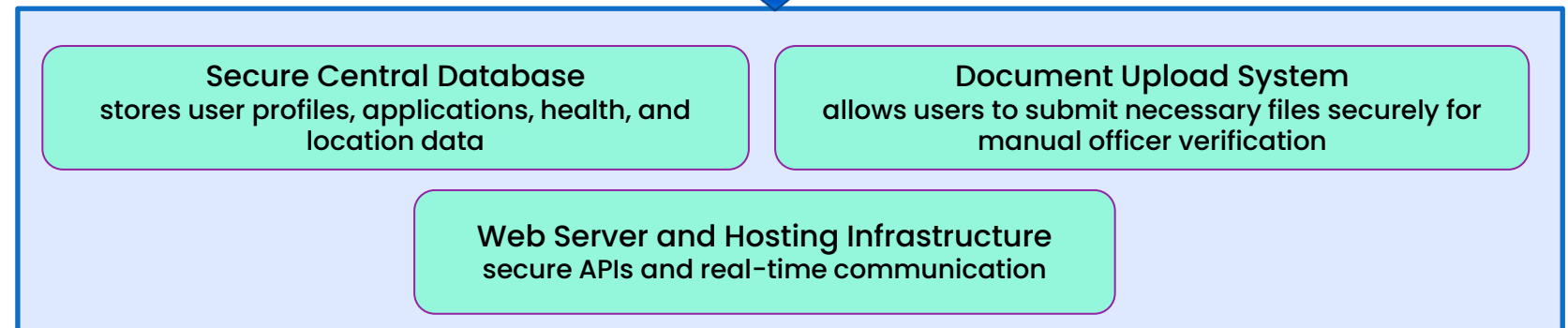
User Interface Layer



AI Intelligence Layer



Infrastructure Layer



“Breaking Barriers in Citizen Welfare Access”



IT22366818 | DISSANAYAKE D.M.V.S

Bachelor of Science (Hons) in Information Technology
Specializing in Information Technology

Student Must add
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photo to this cage

Ai-based Investment Plan Recommendation Module For Digital Welfare Management In Sri Lanka

Research Problem

How can AI-driven clustering and recommender systems help citizens access suitable investments and financial aid schemes, improving financial decision-making and long-term stability?

Lack of personalized financial advice for users

No AI-based recommendations for investments and aid

Limited focus on user risk and income patterns



Research Gap

Aspect	Existing Systems	Gap	Our Solution
Financial Guidance	Generic banking or NGO advisory programs	Not tailored to low-income citizens	AI-driven personalized financial advice
Risk & Income Consideration	Basic income checks only	Limited analysis of risk profiles and spending patterns	AI models analyzing income, expenses, and risk
Accessibility & Scale	Manual counseling, time-consuming	Not scalable for large citizen groups	Automated AI module offering wide accessibility

Previous Research And Shortcomings

Previous Research

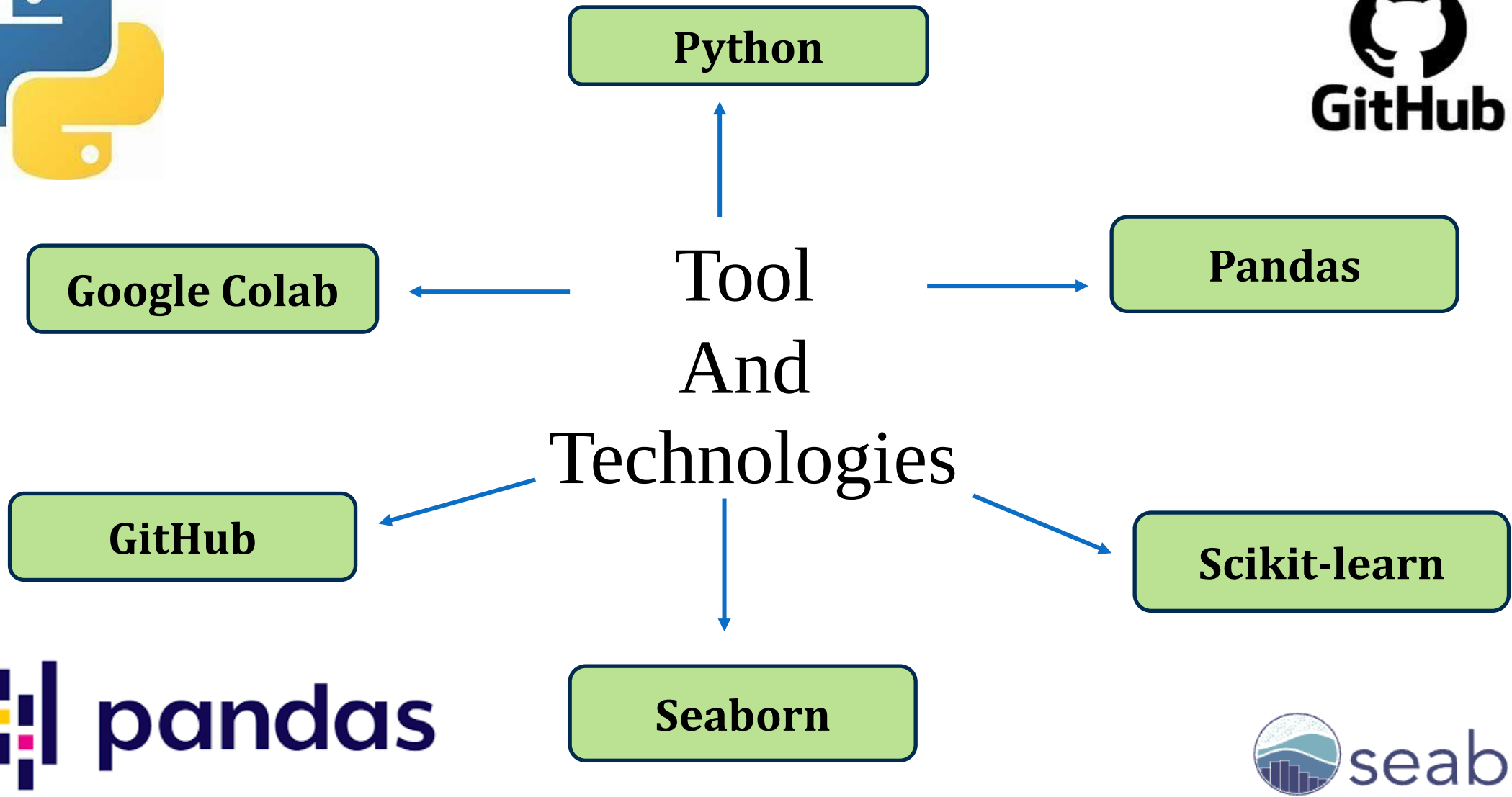
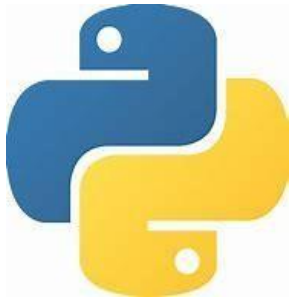
Research on financial inclusion focused on improving literacy and expanding banking access.

FinTech research applied AI to fraud detection, credit scoring, and market predictions.

Shortcomings

Aid platforms give generic advice, ignoring income, expenses, or risks.

FinTech apps often overlook AI support for low-income investment planning.



Methodology

1. Data Collection

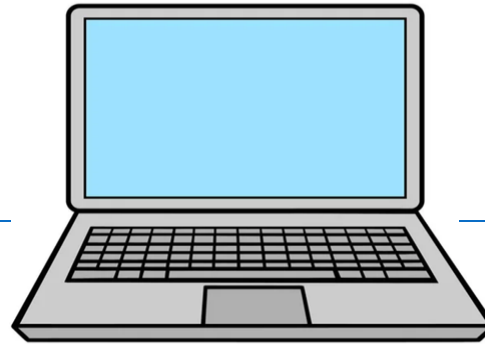
Gather anonymized user income, expenses, and available financial aid schemes.

2. Data Preprocessing

Clean and prepare data for analysis, like handling missing values and normalizing numbers.

3. User Profiling & Clustering

Group users with similar financial behaviors using clustering techniques.



4. Recommendation System

Suggest suitable micro-investments or aid schemes based on user cluster and preferences.

5. Validation & Testing

Check if recommendations are accurate and safe for users.

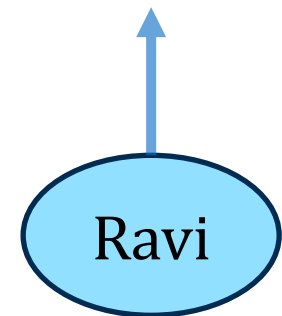
6. Deployment

Make the module usable so users can input data and get recommendations.

A real-world scenario

Ravi is a low-income citizen in Sri Lanka working as a part-time delivery worker. He earns a small monthly income and struggles to save. He wants to invest in a safe micro-investment or apply for financial aid but doesn't know what options suit his income and risk profile.

Using the Investment Plan Recommendation Module, Ravi inputs his monthly income, expenses, and savings. The system analyzes his financial data, clusters him with similar users, and recommends: a micro-savings plan, a low-risk government-backed investment scheme, and an NGO aid program that fits his profile. Ravi can now make safe financial decisions tailored to his situation.



IT22585080 | SHAMIKA A . G

Bachelor of Science (Hons) in Information Technology
Specializing in Information Technology

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Design and Development of a Multilingual Chatbot for Enhancing Accessibility

Research Problem

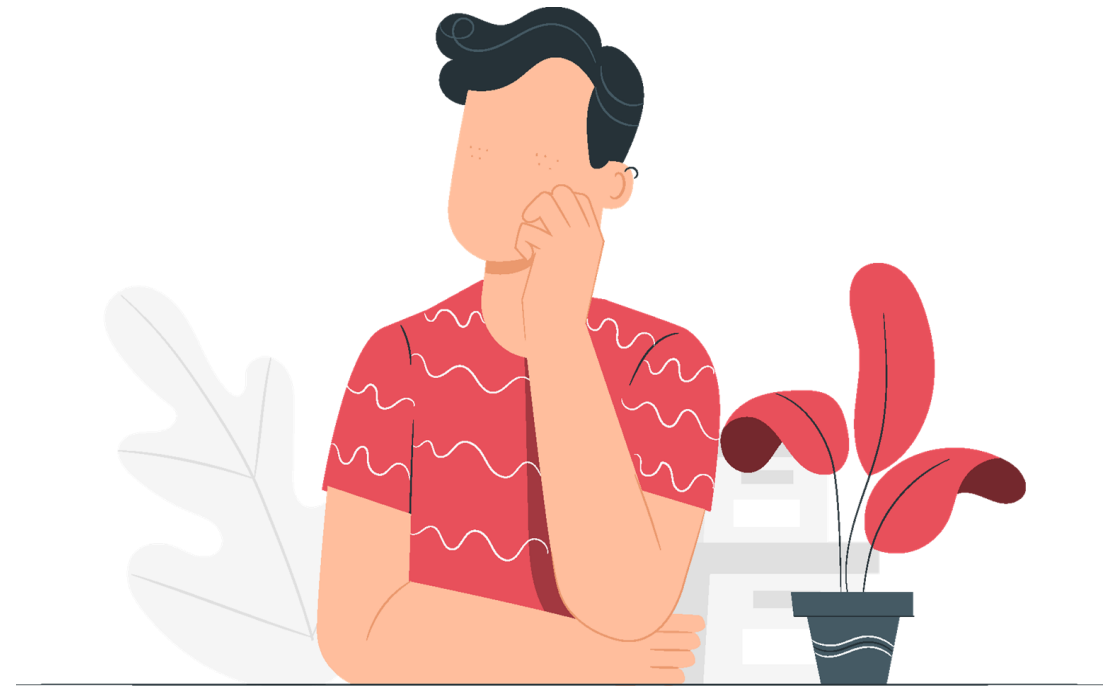
How can a multilingual chatbot be created to accurately process and respond in **Sinhala, Tamil, and English**, considering the linguistic and cultural needs of Sri Lankan welfare recipients?

Language barrier in welfare platforms

Lack of support for Tamil users

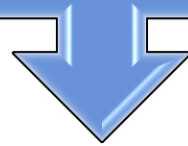
Manual query handling not scalable

Existing bots are rule-based, not intelligent

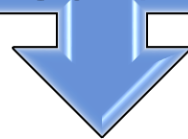


Research Gap

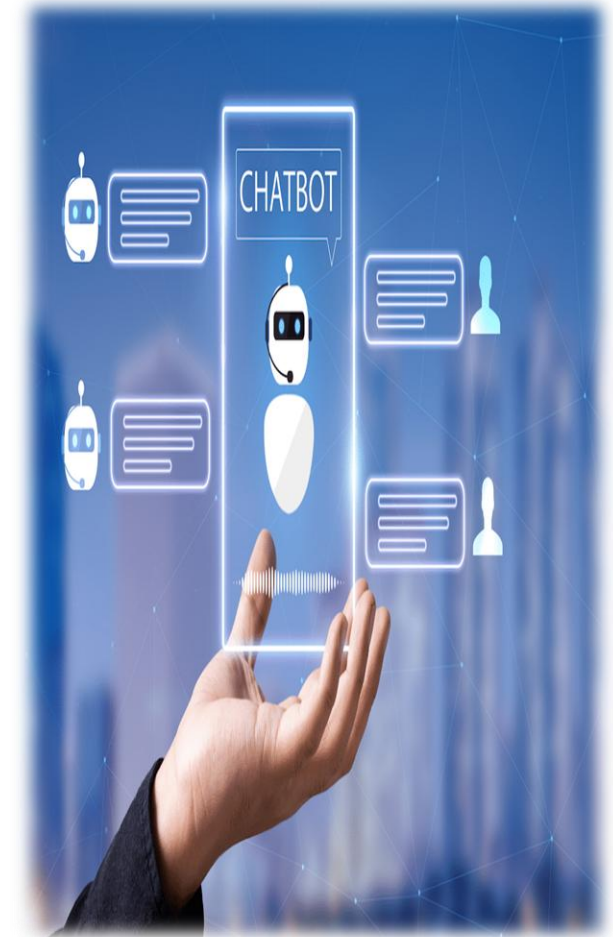
Most welfare service systems in Sri Lanka are **monolingual** (English or Sinhala only).



Limited availability of **trilingual (Sinhala, Tamil, English) chatbots** for government digital services.[2]



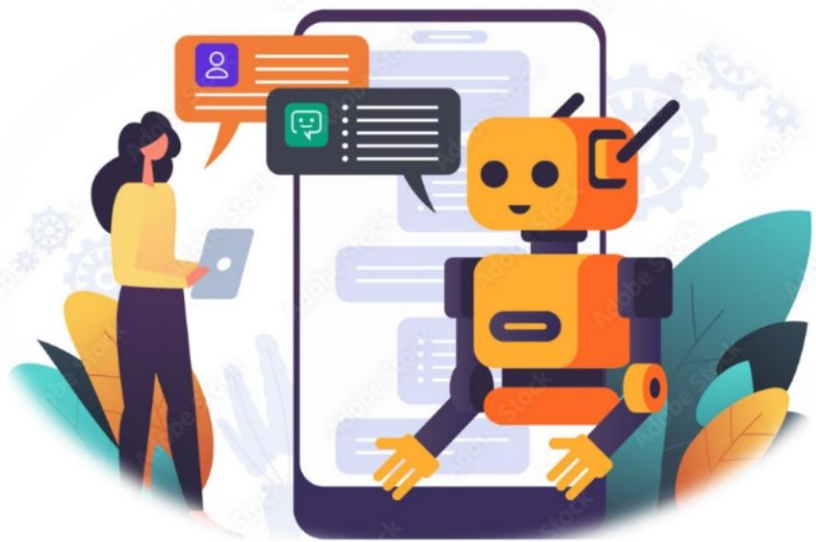
Welfare systems are less accessible for rural and elderly users..[3]



Previous Research and Shortcomings

Previous Research

- Chatbots are common in healthcare, education, and banking, but rarely in welfare services
- Most multilingual research covers global languages, not Sinhala/Tamil.



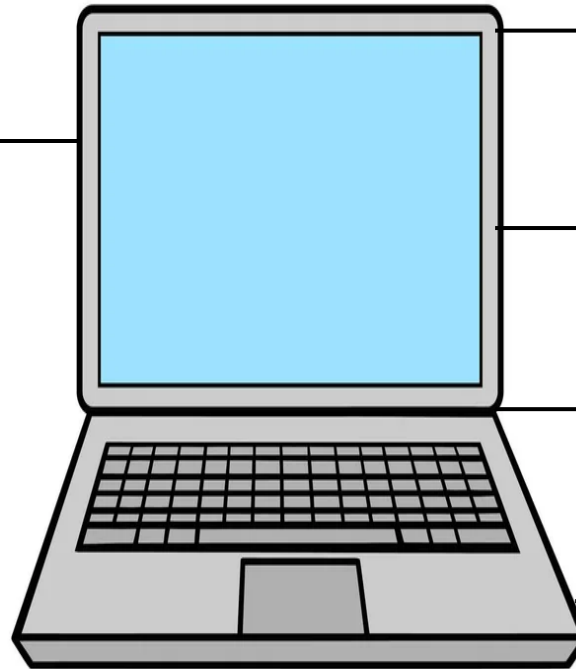
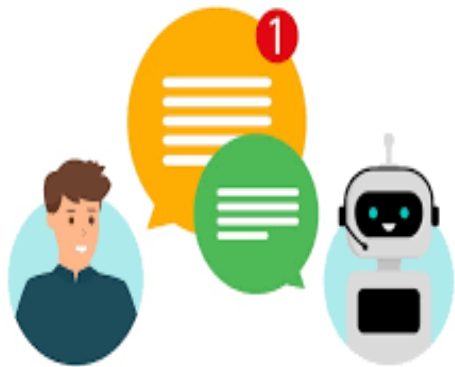
Shortcomings

- Poor **language translation accuracy** (Sinhala/Tamil not well-supported).
- Chatbots use only keyword matching and are not linked with government databases

Specific and Sub Objectives

Specific Objectives

To design and develop an **AI-powered multilingual chatbot** that improves **accessibility, inclusivity, and transparency** in Sri Lanka's welfare services.



Sub Objectives

NLP models for Sinhala, Tamil & English

User-friendly chatbot interface

Integration with welfare systems

Continuous improvement with feedback



Visual Studio
code



MongoDB

Tool And Technologies

Python

Google's
Dialogflow

GitHub



REST APIs

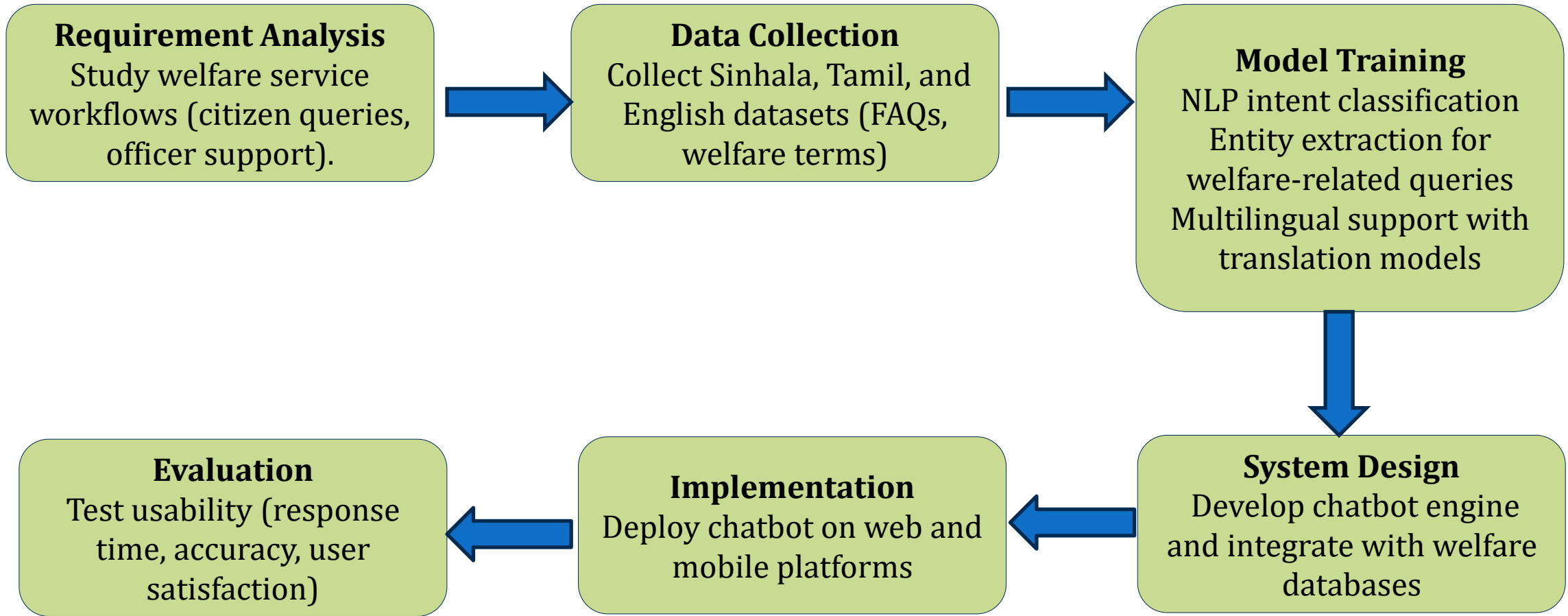


Libraries

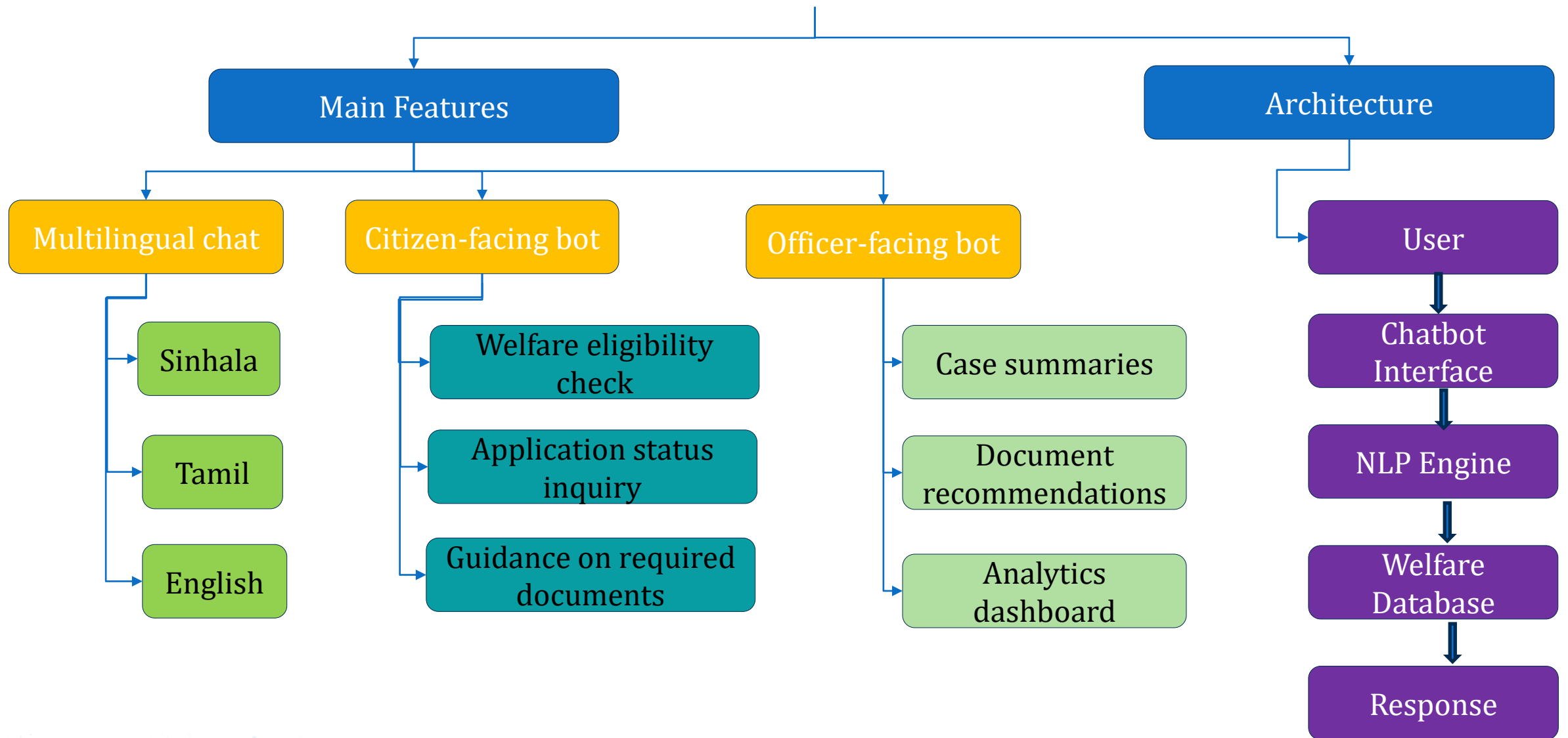
Pandas Library



Methodology



System Overview



Real World Use Case

Citizen Example



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Officer Example



Show me pending
applications in Tamil
Nadu division



Generates a filtered report
with pending applications

Completion of the project

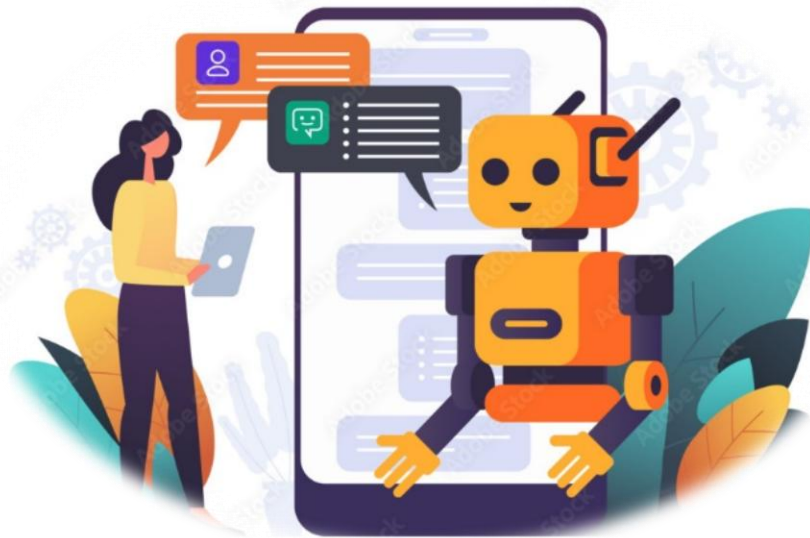
Multilingual chatbot (Sinhala, Tamil, English) for Sri Lanka's welfare services.

Citizen-facing & officer-facing features for better service delivery.

Improved accessibility for rural and elderly users

Integrated with welfare databases for real-time responses and Tested for usability, accuracy, and speed

Enhances inclusivity, efficiency, and transparency



IT22214584 | WICHRAMASINGHE R.P.Y.N

Bachelor of Science (Hons) in Information Technology
Specializing in Information Technology

Student Must add
a professional
photo to this cage

AI-GIS Integrated Vulnerable Population Monitoring and Aid Recommendation System for Welfare Services

Research Problem

How can an AI–GIS integrated system be designed to accurately map vulnerable populations and provide data-driven welfare aid recommendations in Sri Lanka, ensuring fairness, efficiency, and transparency in social service delivery?

Which communities in Colombo District are most in need of welfare support?

How can socio-economic and hazard data be combined into a Social Welfare Need Index (SWNI)?

Can AI and optimization techniques improve the targeting and distribution of welfare aid?



Research Gap

Welfare identification in Sri Lanka is manual, fragmented, and paper-based..

Current systems lack GIS integration → no spatial prioritization of vulnerable areas.

Data silos between census, social services, and disaster management → inefficient planning.

No predictive analytics for anticipating needs during disasters or economic shocks.

Aid allocation lacks fairness checks, leading to under-served communities.

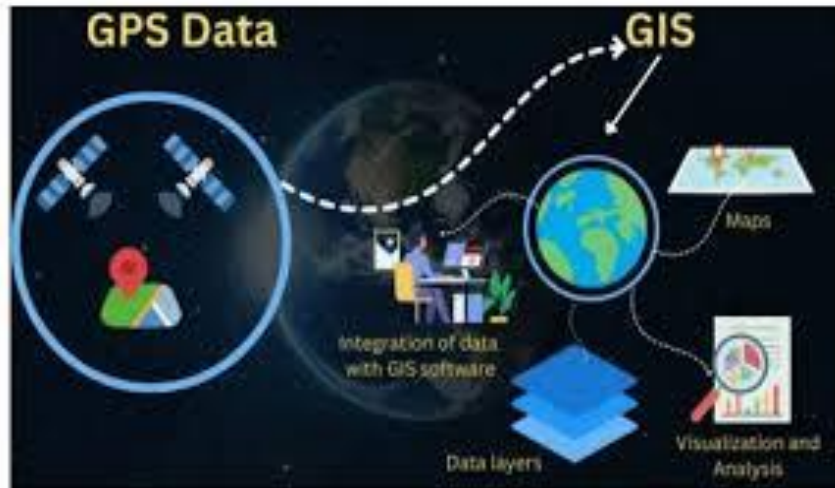


Previous Research and Shortcomings

Previous Research

Sri Lankan welfare (Samurdhi, Aswesuma, disaster relief, GIS) → fragmented, poverty-targeted but with errors; limited socio-economic integration..

Global AI in welfare → predicts poverty but lacks ethics, transparency, and local adaptation.



Shortcomings

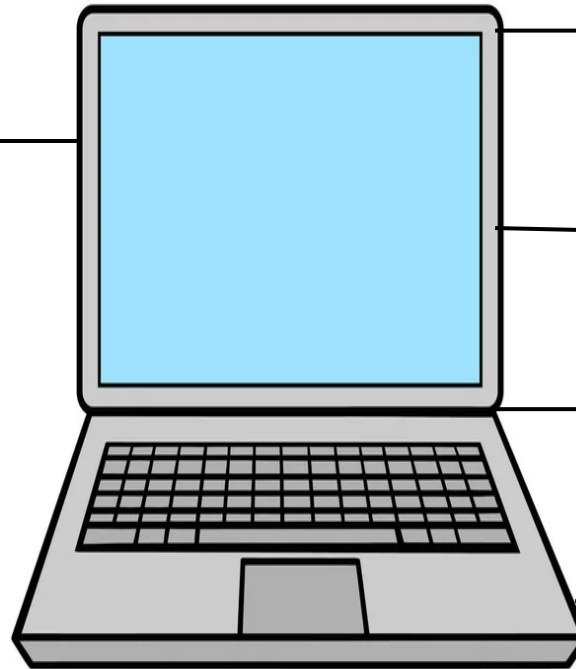
No real-time data & weak accountability in aid delivery

No unified AI-GIS framework for welfare decisions

Specific and Sub Objectives

Specific Objectives

To design and implement an AI-GIS integrated system that identifies vulnerable populations and recommends optimized welfare aid distribution.



Sub Objectives

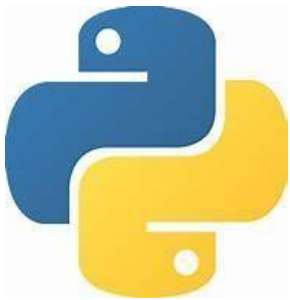
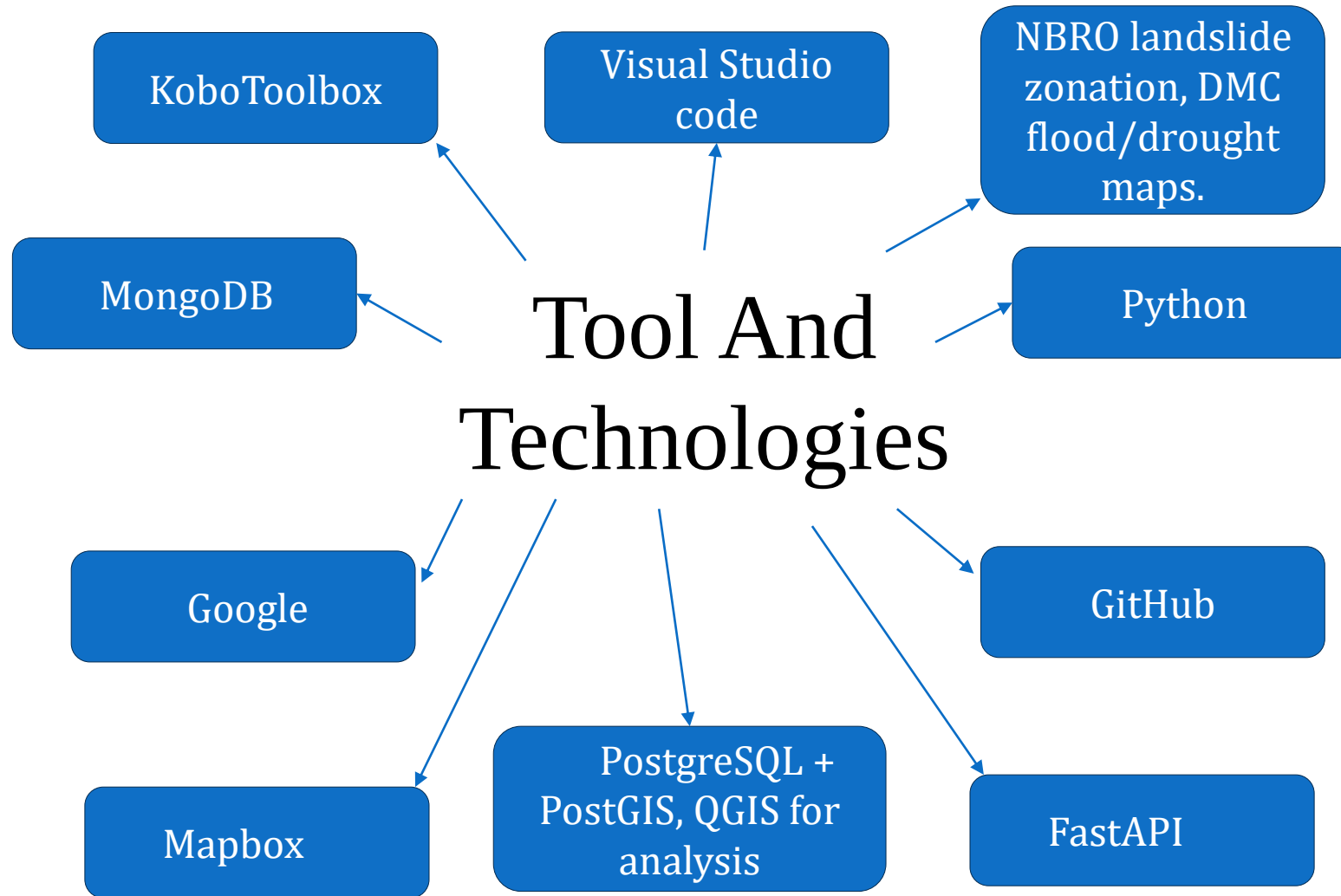
Develop a geospatial database combining welfare, demographic, and hazard data.

Build a Social Welfare Need Index (SWNI) to rank vulnerability at GN/DS level.

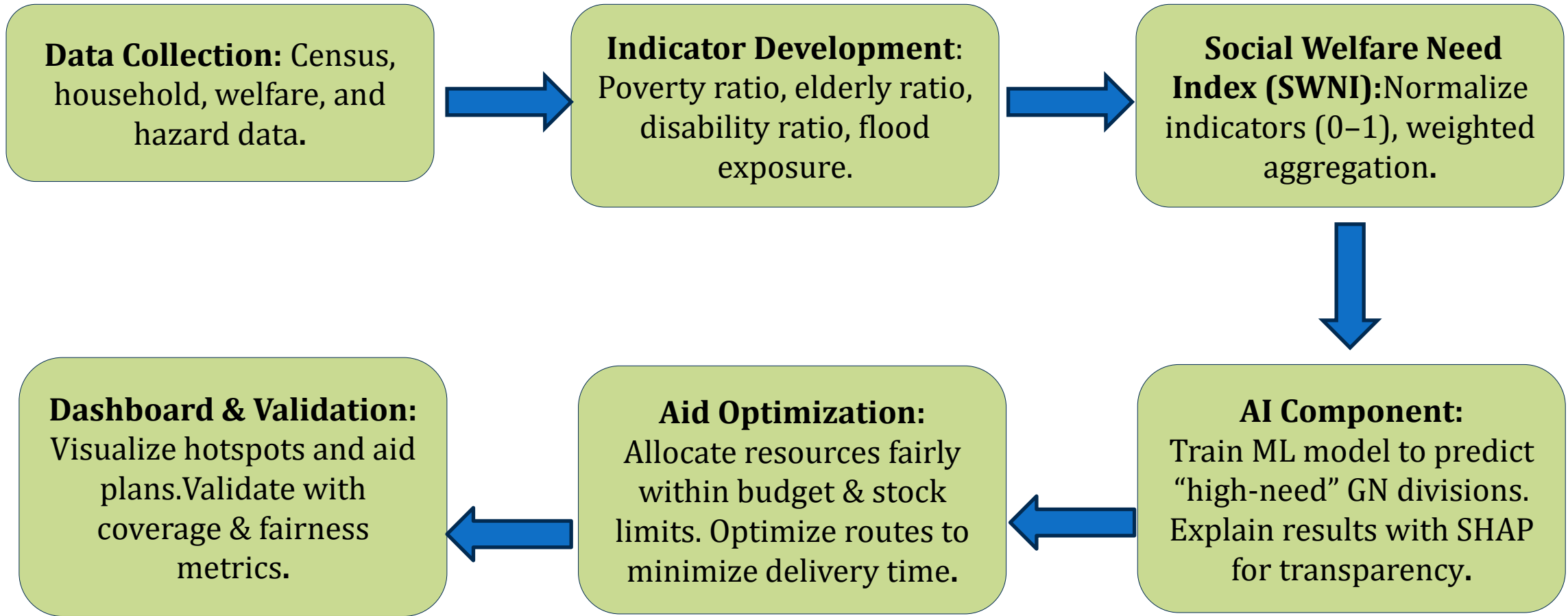
Integrate AI models to predict welfare needs and prioritize groups.

Optimize aid allocation & routing considering budget, stock, and accessibility.

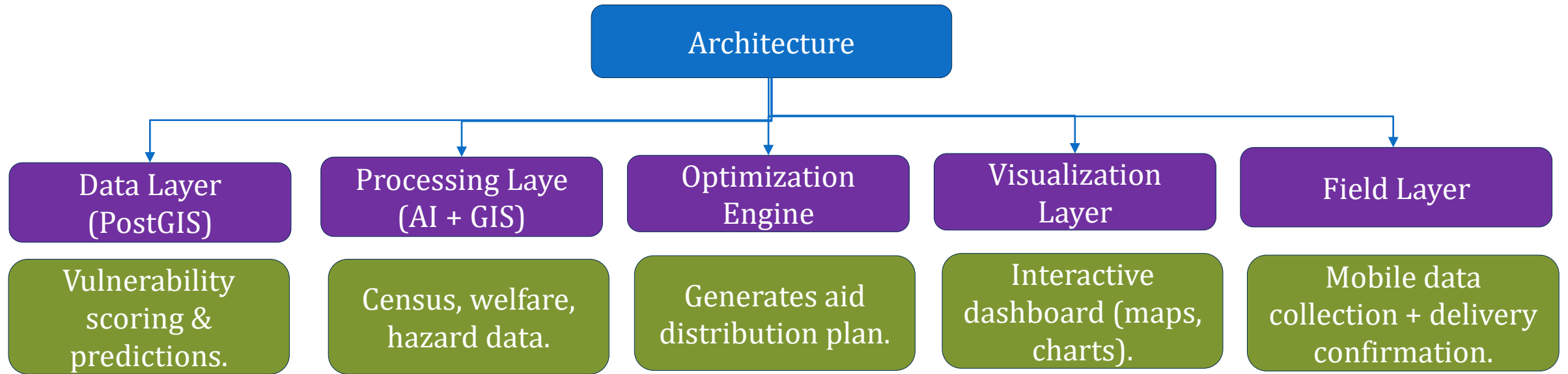
Tool And Technologies



Methodology



System Overview



Output

GN-level SWNI maps.

Priority ranking of communities..

Optimized aid plans with routes.



Real World Use Case

Flood in Kolonnawa DS Division (Colombo)

System highlights GN divisions at highest risk (e.g., Sedawatta, Meethotamulla).

Welfare & hazard data combined → High SWNI zones flagged red.

AI model predicts extra households needing support.

Optimizer generates aid distribution plan:

200 dry-ration packs → 3 GN divisions.

2 routes optimized from main warehouse (time reduced by 30%).

Field officers receive assignment on mobile app → confirm delivery → update system .

Impact: Faster, fairer, data-driven welfare response.



Completion of the project

Data Integration: Combined census, welfare, and hazard datasets into a PostGIS spatial database with privacy safeguards

Vulnerability Mapping: Built a Social Welfare Need Index (SWNI) and produced GN-level heatmaps..

Outcomes: A scalable, ethical AI-GIS platform improving fairness, efficiency, and transparency in welfare services..



AI Component: Applied ML models (XGBoost, Logistic Regression) + Explainable AI (SHAP) for predicting vulnerable areas

Aid Optimization: Developed an allocation & routing engine (OR-Tools, OSRM) for fair and efficient distribution.

System Prototype: Built a dashboard (FastAPI + React/Leaflet) with maps, rankings, and aid plans

IT22926494 | MANAWADU G M S T

Bachelor of Science (Hons) in Information Technology
Specializing in Information Technology

Student Must add
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AI-Driven Eligibility Prediction and Healthcare Recommendation System

Research Problem & Research Gap

How can Artificial Intelligence be used to fairly and accurately predict welfare eligibility while recommending relevant healthcare services tailored to citizen profiles in Sri Lanka?

Aspect	Existing Systems	Gap	Our Solution
Welfare Eligibility	Manual verification in Sri Lanka, “Aadhaar” in India	No AI-based prediction	AI eligibility model for fast & fair decisions
Healthcare Integration	Welfare systems in India & Bangladesh	No healthcare support linked to welfare	Add healthcare recommendation engine
Accuracy & Personalization	Basic digital records only	Errors, no tailored support	ML models + personalized health referrals

Previous Research And Shortcomings

Existing Research

(Global Work)

- **Welfare Fraud Detection**
 - AI systems in **India & USA** to reduce fraud in welfare distribution.
- **Predictive Analytics in Healthcare**
 - Used by **WHO & OECD** for resource allocation.
- **Poverty & Income Classification**
 - **World Bank** studies applying ML to socio-economic data.
- **Recommendation Systems in Health**
 - Rule-based referral systems tested in developed nations.

Shortcomings & Our Gap

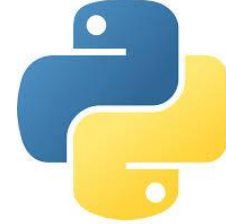
(Gaps for Sri Lanka)

- **No Localized Data**
 - Global models ignore Sri Lankan socio-economic and healthcare context
- **Narrow Focus**
 - Most target only **eligibility**, not healthcare integration.
- **“Black-Box” AI**
 - Citizens & officers can’t see why eligibility was granted/denied.
- **Weak Government Integration**
 - Few solutions link directly with **welfare portals & healthcare networks**.
- **Fairness & Trust Issues**
 - Existing AI models raise bias and transparency concerns.

Tool And Technologies

Development & Environment

- Visual Studio Code → IDE for coding
- Python → Core programming language
- GitHub → Version control & collaboration



Machine Learning & Data

- Pandas, NumPy → Data preprocessing & cleaning
- Scikit-learn → Decision Trees, Logistic Regression



Integration & Deployment

- REST APIs → Connect eligibility + healthcare modules to welfare portal
- Flask → Backend frameworks
- MongoDB → Database for socio-economic + healthcare data



Other Services (if extended)

- Google Cloud Services → Hosting, storage, ML deployment
- Google Maps → For healthcare location recommendations



Methodology

1. Data Collection:

Socio-economic and healthcare datasets from **Sri Lanka Open Data, WHO Health Indicators, HIES 2019, Kaggle income datasets**.

2. Data Preprocessing:

Handle missing values, normalize numerical data, encode categorical variables. Normalize numerical features

3. Model Training:

Split data into **training and testing sets (80:20)**.

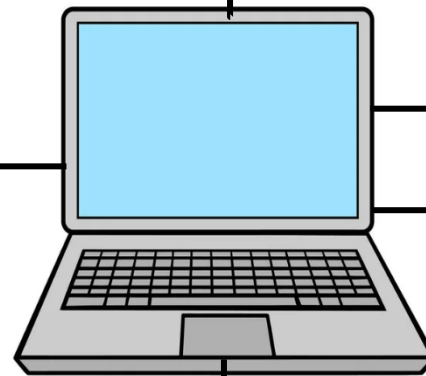
Train **Decision Tree** and **Logistic Regression** models to classify eligibility (Python Scikit-learn, TensorFlow). Validate with **accuracy, precision, recall, and cross-validation**.

4. Healthcare Recommendation Engine:

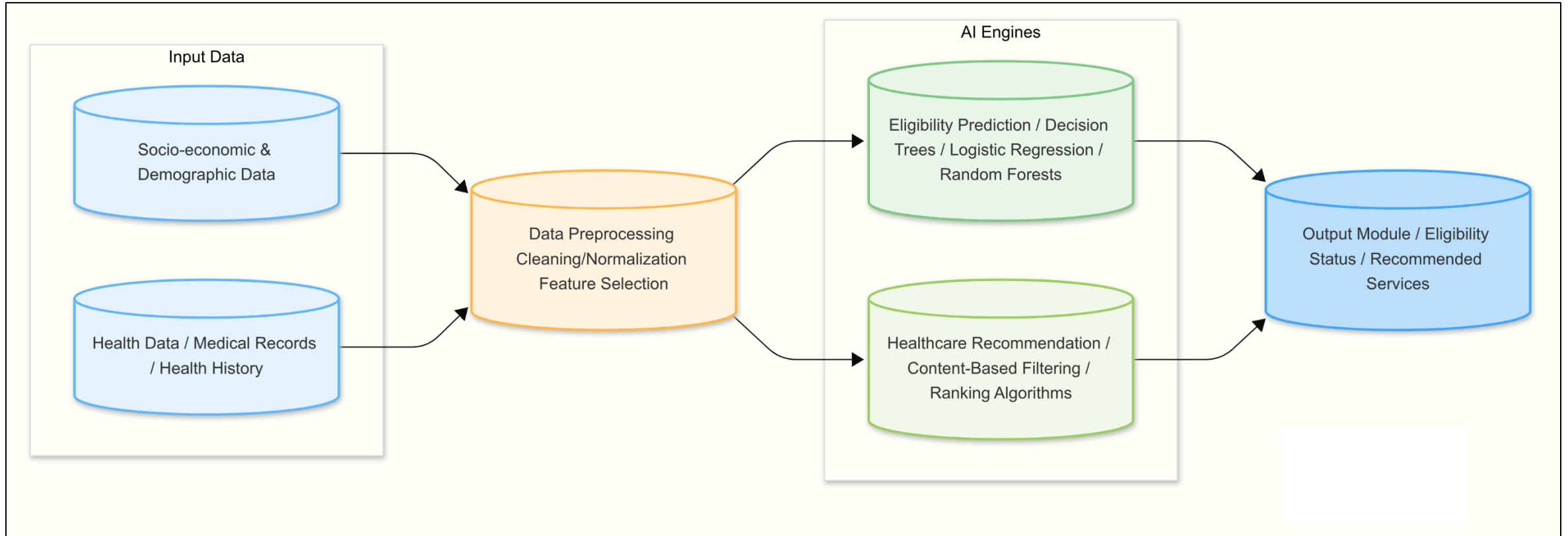
Rule-based + similarity-based algorithms suggest hospitals or specialists based on citizen health profiles.

5. Integration:

Connect the AI modules to the welfare portal via **APIs**, enabling officers to view eligibility and recommendations in real-time



Individual System Diagram



A Real-world Scenario: A Low-Income Family in Rural Sri Lanka



Profile Card:

Family Size: 5

Monthly Income: LKR 15,000

Age: 42

Health Condition:

Hypertension

System Prediction:

Welfare Eligibility: Eligible for Samurdhi Program

Healthcare Recommendation:

Nearby government clinic for monthly free checkups

Impact:

Faster Approval (for the citizen)

Holistic Support

Time Saved (for the government officers)

Let's make this concrete. Meet a 42-year-old mother of three with hypertension. Our system doesn't just say 'Yes' to welfare. It also proactively recommends the specific healthcare services she needs, like free monthly checkups at her local clinic. This saves time for officers and provides holistic support for citizens

Investment / Cost

Development Costs – Software, AI models, integration

Infrastructure Costs – Cloud hosting, servers, database security

Human Resources – Developers, data scientists, domain experts

Maintenance & Updates – Continuous support, bug fixes

Description	Cost	Occurring
Server Hosting (Cloud VPS)	LKR 10,000	Monthly
Cloud Database Storage (Amazon S3 / Firebase)	USD 0.023 per GB	Monthly
Domain Registration & SSL Certificate	LKR 5,000	Annual
Third-Party API Costs (Health Data, AI Models)	LKR 15,000	Monthly
Marketing & Promotion	LKR 50,000	Initial Launch
Maintenance & Technical Support	LKR 25,000	Monthly

Ability of Commercialization & Future Prospects

Target Market:

Governments, healthcare providers, NGOs (Main client)

- Position as a national welfare platform for the Sri Lankan government.
- Provide specialized modules for NGOs to manage targeted welfare programs.

Researchers & Private Companies (Data Access)

- Offer secure, anonymized datasets to universities, health researchers, and private sector organizations.
- Support evidence-based policymaking, academic studies, and market analytics while ensuring data privacy.

Advertisement Revenue on Platform

- Allow ethical, welfare-related advertisements (e.g., healthcare services, educational programs, insurance).
- Generate continuous revenue without affecting citizen services.

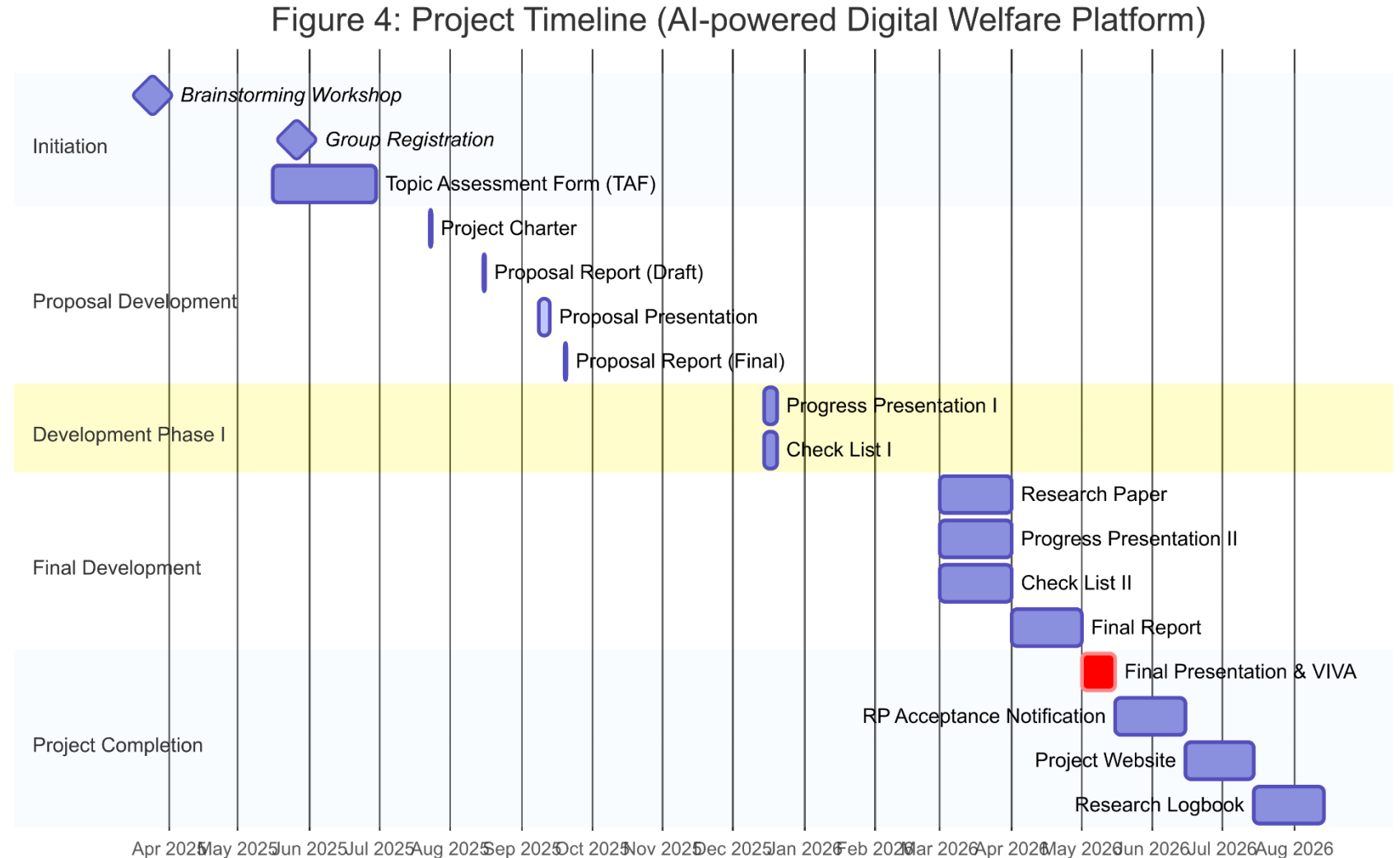
Future Prospects (Subscription Features)

Expand regionally across South Asia for broader welfare management.

Introduce subscription-based advanced features

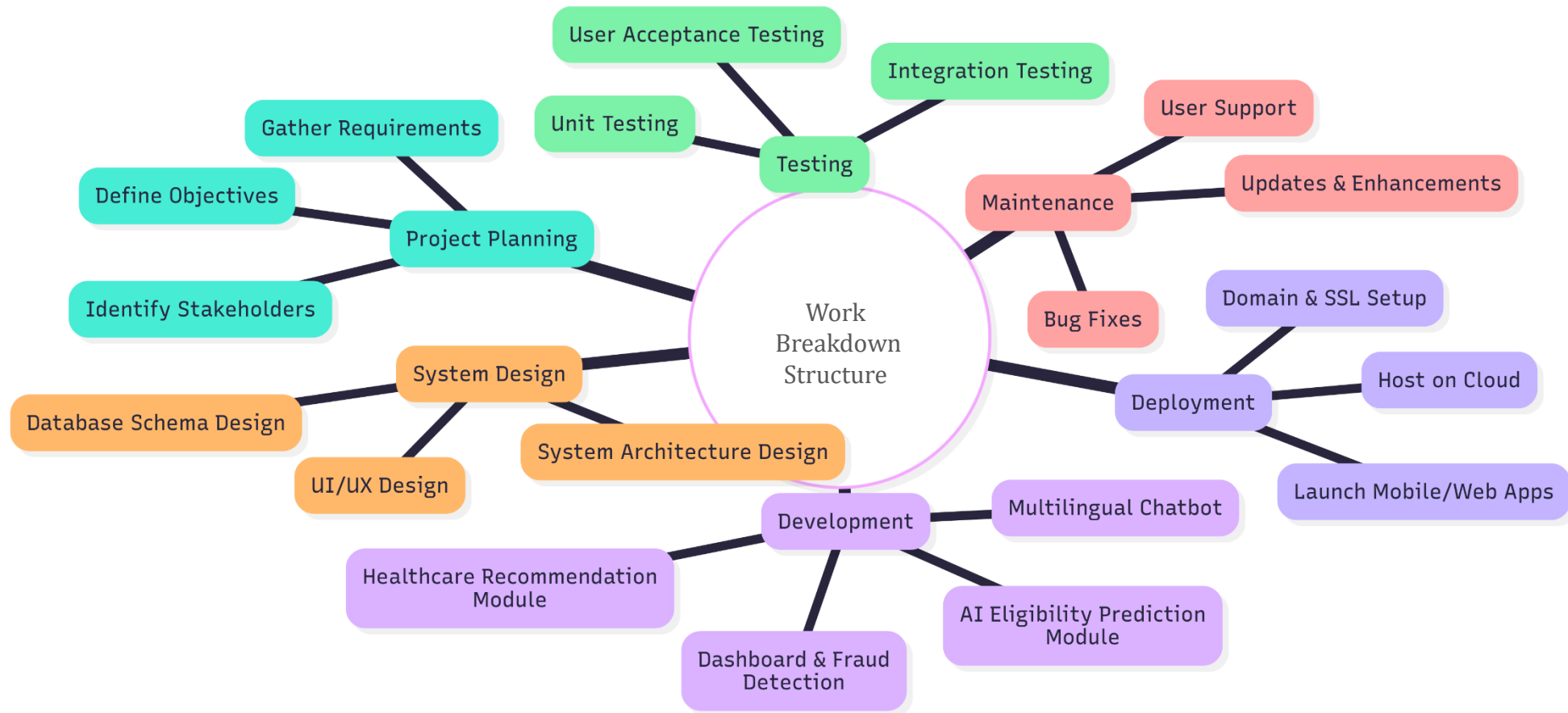
Completion of the project

Gantt Chart



Completion of the project

Work Breakdown Structure



References

- [1] M. U. Ahmed, AI Applications in Social Assistance Programs, South Asian Journal of Policy and Governance, 2019.
- [2] S. Bhatnagar, E-Government: From Vision to Implementation. SAGE Publications, 2004.
- [3] World Bank. (2021). Sri Lanka Poverty Assessment. Washington, DC: World Bank.
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- [5] M. U. Ahmed, "Digital transformation of social welfare services: Challenges and opportunities in South Asia," South Asian Journal of Policy and Governance, 2019.
- [6] United Nations, E-Government Survey 2022: Digital government in the decade of action for sustainable development. New York, NY, USA: United Nations, 2022.
- [7] M. U. Ahmed, "AI Applications in Social Assistance Programs," South Asian Journal of Policy and Governance, 2019.
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